GREENIE

Fleet Intelligence + Marketplace

TECHNICAL DESIGN DOCUMENT

Marketplace Module — Architecture, UI/UX & Integration Spec

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Version	Status	Date	Relates to	City
1.0	**Draft**	**Mar 2026**	**PRD v1.0**	**Coimbatore**

1. Document Overview

This Design Document specifies the technical architecture, database schema, API design, UI/UX system, screen specifications, and integration blueprint for the GREENIE Marketplace Module. It is read alongside the GREENIE Marketplace PRD v1.0 and assumes familiarity with the existing GREENIE Fleet Intelligence system (harishsankarks.github.io/Greenie-Fleet-Intelligence).

2. Full System Architecture

2.1 Two-System Overview

GREENIE now operates two interconnected systems. The Fleet Intelligence System (existing) manages inbound waste, fleet routing, and Transfer Station operations. The Marketplace Module (new) manages outbound material sales. The two systems share a single admin identity and communicate via a lightweight API bridge.

FLEET INTELLIGENCE (Existing) *harishsankarks.github.io/Greenie-Fleet-Intelligence*	MARKETPLACE MODULE (New) *marketplace.greenie.ac.in*
• Contractor Requests & Tipping Fees • Autonomous Fleet GPS Routing (DRL/CVRP) • Transfer Station Edge AI Sorting (ESP32 + CNN) • Sorting Batch Log & Weight Sensor Data • Admin Login: 123@greenie.ac.in	• Buyer Registration & KYC • Material Catalog (inventory auto-synced) • Cart, Checkout & Payment

• Client Portal (Contractor access)	• Order Tracking (linked to fleet dispatch) • WTN + EPR Certificate Generation • Admin: same SSO as Fleet Intel • Analytics & Pricing Dashboard
API BRIDGE: Fleet Intel → Marketplace | Webhook events: SortingComplete, FleetDispatched, DeliveryConfirmed	

2.2 Technology Stack — Marketplace Module

Layer	Technology	Rationale
Frontend	React 18 + Next.js 14 (SSR/ISR)	SEO-friendly material catalog; fast load on low-bandwidth field devices; ISR for real-time stock updates
Mobile	PWA (Progressive Web App)	Works on Android phones at Transfer Stations and field; no app store required for Phase 1
Backend API	Node.js + Express (REST)	Consistent with common GREENIE tech; rapid development; easy Fleet Intel integration
Primary Database	PostgreSQL 15	ACID transactions for orders; JSONB for flexible material metadata; PostGIS for zone proximity
Cache	Redis 7	Inventory cache (5 min TTL); session management; notification rate limiting
Search	Elasticsearch 8	Material search with grade filters and zone-distance ranking
File Storage	AWS S3 + CloudFront	Transfer Station batch photos, lab certificates, WTN/EPR PDFs
Payments	Razorpay	UPI, Net Banking, NEFT; PO invoice generation; escrow for large orders
PDF Generation	Puppeteer (Node.js)	WTN + EPR + Tax Invoice PDF rendering from templates
Hosting	Vercel (Frontend) + AWS ECS (Backend)	Low-cost, GitHub-deploy-friendly for college project scale; production-ready architecture
Auth	JWT + same credentials as Fleet Intel	Admin SSO: 123@greenie.ac.in works across both systems

3. Database Schema

3.1 Core Tables

Table	Key Columns	Notes
buyers	id, name, phone (OTP-verified), email, gstin, buyer_type (municipal/recycler/contractor/individual), kyc_status, created_at	Separate from contractors in Fleet Intel; buyer KYC managed here
material_inventory	id, source_station (N/S/E/W), material_type, grade, quantity_kg, reserved_kg, price_per_unit, unit, lot_photos[], lab_cert_url, batch_ref (FK to Fleet Intel batch), status (available/reserved/sold), synced_at	Auto-created via webhook from Fleet Intel sorting event
listings	id, inventory_id, title, description, min_order_qty, published_at, expires_at	GREENIE admin publishes from inventory draft; thin layer over inventory
orders	id, buyer_id, status, total_amount, delivery_mode (pickup/delivery), delivery_slot, station_id, gst_amount, invoice_url, wtn_url, epr_cert_url, created_at	Core order record linking buyer to material
order_items	id, order_id, inventory_id, quantity_kg, unit_price, subtotal	Line items; quantity reserved in inventory on order creation
payments	id, order_id, razorpay_order_id, razorpay_payment_id, amount, status, paid_at	Razorpay payment record
reviews	id, buyer_id, order_id, material_quality (1-5), grade_accuracy (1-5), delivery_punctuality (1-5), comment, admin_reply	Post-delivery review from buyer
fleet_events	id, event_type (SortingComplete/Dispatched/Delivered), fleet_intel_ref, order_id, payload_json, received_at	Webhook events received from Fleet Intel; processed async
epr_certificates	id, order_id, cert_number, buyer_gstin, material_type, quantity_kg, issued_at, pdf_url, blockchain_hash	Auto-generated on order completion; hash stored for CPCB audit
wtn_records	id, order_id, wtn_number, generator (GREENIE), recipient_id, material_type, quantity_kg, issued_at, pdf_url	Waste Transfer Note per transaction

4. API Design

4.1 Marketplace REST API

Method + Endpoint	Description	Auth
GET /catalog	List all available material listings with filters (material, grade, zone, qty range, price range)	Public
GET /catalog/:id	Get material listing detail with photos, cert, zone map, stock qty	Public
POST /buyers/register	Register new buyer (name, phone OTP, type, GSTIN)	Public
POST /buyers/kyc	Upload KYC documents; triggers admin review queue	Buyer JWT
POST /cart	Create/update cart with items and quantities	Buyer JWT
POST /orders	Create order from cart; reserve inventory; generate Razorpay order ID	Buyer JWT
GET /orders/:id	Order detail with status timeline	Buyer / Admin
POST /payments/verify	Verify Razorpay payment signature; confirm order	Buyer JWT
POST /orders/:id/confirm-receipt	Buyer confirms delivery received; triggers WTN + EPR cert generation	Buyer JWT
POST /reviews	Submit post-delivery review	Buyer JWT
GET /admin/inventory	Admin: view all inventory batches, stock levels per station	Admin JWT
PUT /admin/inventory/:id/publish	Admin: publish listing from inventory draft	Admin JWT
PUT /admin/pricing/:material	Admin: set/update price per material and grade	Admin JWT
GET /admin/orders	Admin: all orders with filter/search	Admin JWT
GET /admin/analytics	Revenue, GMV, turnover per material/zone, price trend	Admin JWT

4.2 Fleet Intelligence Integration Webhooks

Webhook Event	Payload	Marketplace Action
POST /webhooks/sorting-complete	station_id, material_type, grade, quantity_kg, batch_ref, lot_photos[]	Create material_inventory record (status: draft); notify admin to review & publish
POST /webhooks/fleet-dispatched	order_id (marketplace), truck_id, driver_name, driver_phone, gps_tracking_url	Update order status → In Transit; send buyer SMS with tracking link
POST /webhooks/delivery-confirmed	order_id, driver_photo_url, delivered_at	Update order → Delivered; trigger WTN + EPR PDF generation; prompt buyer to review

5. Information Architecture

PUBLIC	BUYER PORTAL	GREENIE ADMIN
• Homepage / Hero • Material Catalog • Product Detail • How to Buy • Impact Counter • Register / Login • About GREENIE	• Dashboard • Browse & Search • Product Detail • Cart & Checkout • My Orders + Tracking • Documents (WTN, EPR, Invoice) • Reviews • Account / KYC Status	• Overview Dashboard • Inventory Manager • Pricing Control • Order Management • Buyer KYC Queue • Document Centre (WTN / EPR) • Analytics & Reports • Fleet Intel Link

6. UI/UX Design System

6.1 Design Principles

- GREENIE Brand Fidelity: Green + Amber (yellow) color scheme directly from GREENIE visual identity. No deviation.
- Single-Seller Simplicity: No complex seller profiles, no 'compare sellers'. Product page shows GREENIE's certified material directly.
- Compliance-Forward: WTN and EPR certificate download are first-class features — not buried in settings. Buyers come for material AND documentation.
- Field-Ready: Large touch targets, minimal forms, Tamil language toggle for local contractors and individuals in Coimbatore.

6.2 Color Palette

Token	Hex	Usage
--greenie-primary	#1B5E20	Primary CTAs, headers, nav bar — GREENIE deep green
--greenie-primary-mid	#2E7D32	Button hover, active states, section headers
--greenie-primary-light	#E8F5E9	Page backgrounds, card fills, table alternates
--greenie-amber	#F9A825	Accent highlights, badges, 'New Stock' chips — GREENIE yellow
--greenie-amber-dark	#E65100	Alert banners, urgent stock indicators, key callouts
--greenie-amber-light	#FFF9C4	Info boxes, compliance callouts, admin notes
--text-primary	#212121	All body text, headings
--text-secondary	#616161	Labels, meta, placeholders

Token	Hex	Usage
--white	#FFFFFF	Cards, modals, surfaces
--border	#BDBDBD	Table borders, input outlines, dividers

6.3 Typography

Style	Font	Size	Weight	Usage
Display	Inter / Noto Sans Tamil	32px	700	Hero headline: 'Sorted. Certified. Delivered.'
H1	Inter	24px	700	Page titles
H2	Inter	20px	600	Section headings, card titles
Body	Inter	16px	400	Descriptions, detail text
Small	Inter	14px	400	Meta: zone, date, grade label
Caption	Inter	12px	400	Cert footnotes, legal disclaimers
Tamil Body	Noto Sans Tamil	16px	400	Tamil UI toggle for local buyers

6.4 Key UI Components

Component	Variants	Behaviour
MaterialCard	Grid / List	Photo (4:3), Grade badge (amber chip), Price/unit, Zone tag (N/S/E/W), Stock qty, 'Add to Cart' button. Amber 'LOW STOCK' banner when <5 tonnes.
ZoneFilter	Horizontal tabs	Tabs: All / North / South / East / West Transfer Station. Each tab shows live stock count.
StockTicker	Homepage widget	Animated real-time counters: 'X tonnes available now' per material type. Driven by inventory API.
CheckoutStepper	3 steps	Step 1: Bag Review + Zone. Step 2: Delivery Mode (Pickup / GREENIE Delivery). Step 3: Payment + GST. Progress bar top.
DocumentCard	WTN / EPR / Invoice	Green badge icon, document type label, download button, blockchain hash (truncated). Available in My Orders.
OrderTimeline	Vertical stepper	Confirmed → Preparing at Station → Loaded → In Transit (live GPS) → Delivered → Review. Fleet Intel events auto-update.
AdminInventoryRow	Table row	Batch ref, Material, Grade, Station, Qty, Status (Draft/Live/Sold), Price, Edit/Publish buttons.
ImpactBanner	Homepage section	Cumulative: Tonnes Diverted, CO2 Saved, EPR Certs Issued — animated green counters.

7. Key Screen Specifications

7.1 Homepage

Nav Bar	Hero Section	Material Grid	Impact Counter
GREENIE logo + Marketplace Material category icons Sign In / Register	'Sorted. Certified. Delivered.' Live stock ticker Browse Material CTA	5 category cards Stock qty live → filtered catalog	Tonnes diverted CO2 saved EPR certs issued

7.2 Material Catalog Page

- Left panel (25% width, drawer on mobile): Zone tabs (All / N / S / E / W), Material type checkboxes, Grade multi-select, Quantity range slider, Price range slider, Sort by (Price, Stock qty, Nearest zone).
- Right panel (75%): Grid of MaterialCards; 3-column desktop, 1-column mobile. Active filter chips above result count.
- Inline stock count update every 60 seconds via SSE (Server-Sent Events) — no full page reload.

7.3 Material Detail Page

LEFT — Photos	RIGHT — Purchase Panel
• Swipeable lot photos from Transfer Station • Grade badge + GREENIE Verified tick • Edge AI Sorted certification badge • Enlargeable on tap	• Material name + Grade (prominent) • Price/tonne — set by GREENIE admin • Available stock + Min. order qty • Source: [Zone] Transfer Station • Lab cert download (if available) • Quantity input + Add to Cart • Delivery slot picker • WTN + EPR cert included badge

Below the fold: Order Timeline explainer → Buyer Reviews section → Related materials (same zone, different grade).

7.4 Admin Dashboard — GREENIE Staff

Inventory	Today's Orders	Fleet Sync	Compliance
Pending batches Live listings qty Low stock alerts	New orders count Revenue today Pending deliveries	Last webhook received Pending dispatch assign Live trucks	WTNs issued today EPR certs this month CPCB report due

Below: Inventory table (all batches with Draft/Live/Sold status) + Order management table with fleet assignment link to Fleet Intelligence dashboard.

8. Security Design

Area	Control
Admin SSO	Same JWT credentials as Fleet Intelligence (123@greenie.ac.in). Token shared via OAuth2-style exchange — one login for both systems.
Buyer Auth	Phone OTP (Twilio) at registration. JWT access tokens (15 min) + refresh (30 days). Buyer cannot access admin routes.
KYC Data	Aadhaar OTP via DigiLocker; GSTIN validation via GST portal API. PII encrypted AES-256 at rest.
Webhook Security	Fleet Intel → Marketplace webhooks signed with HMAC-SHA256 shared secret. Marketplace rejects unsigned events.
Payment	Razorpay tokenization. No card data on GREENIE servers. Signature verification on payment callback.
RBAC	Two roles: admin (full access) and buyer (catalog, orders, own docs only). Row-level enforcement in PostgreSQL.
Data Privacy	DPDP Act 2023 compliant. Buyer data not shared with third parties. Right to erasure supported for individual buyers.

9. Performance & Reliability

- Next.js ISR: Material listing pages revalidated every 60s — fast load, near-real-time stock. No SSR cost for every buyer request.
- Inventory cache in Redis (5 min TTL). Invalidated immediately when Fleet Intel webhook updates stock.
- Webhook events processed via async queue (Bull + Redis) — Fleet Intel events never blocked by marketplace DB slowness.
- Transfer Station photo pipeline: S3 upload → Lambda resize (thumbnail 400x300) → CloudFront CDN. Buyers on 4G see <500KB page.
- Target: Catalog page <2s on 4G; Checkout <3s; Webhook processing <5s end-to-end.

10. Compliance Document Generation

Document	Trigger	Generation Method	Delivery
GST Tax Invoice	Payment confirmed	Puppeteer PDF from template; IRN fetched from GST e-invoice API	Email + My Orders download
Waste Transfer Note (WTN)	Buyer confirms receipt	Puppeteer PDF; WTN number = GREENIE-{station}-{date}-{seq}; signed by GREENIE admin	My Orders + WhatsApp link
EPR Certificate	WTN issued	PDF from CPCB-compliant template; SHA-256 hash stored in epr_certificates table	My Orders + buyer email
Blockchain Manifest	EPR cert issued	Hash of WTN+EPR+order data written to Polygon testnet (Phase 2 — CPCB portal integration)	Cert QR code links to block explorer

11. Deployment Architecture

Component	Deployment	Notes
Frontend (Next.js)	Vercel (GitHub push → auto deploy)	Free tier sufficient for MVP; custom domain marketplace.greenie.ac.in
Backend API (Node.js)	AWS ECS Fargate (single service, 0.5 vCPU, 1GB)	Auto-scales to 2 instances at >100 concurrent users
PostgreSQL	AWS RDS (db.t3.micro — free tier eligible)	Daily automated backups; Multi-AZ in Phase 3
Redis	AWS ElastiCache (cache.t3.micro)	Session + inventory cache
File Storage	AWS S3 + CloudFront	Standard storage; lifecycle policy archives sold-lot photos after 90 days
CI/CD	GitHub Actions (same repo as Fleet Intel)	Lint → Test → Build → Deploy on merge to main
Domain	marketplace.greenie.ac.in	CNAME to Vercel; SSL via Let's Encrypt auto-renew

12. Open Design Decisions

Decision	Options	Recommended	Due
Tamil language UI	i18next full translation vs Google Translate widget	i18next for product/checkout	Sprint 3

Decision	Options	Recommended	Due
		pages; auto-translate for blog	
Blockchain for EPR manifest	Polygon testnet vs Hyperledger vs just PDF hash	Polygon testnet (low-cost, CPCB-friendly narrative)	Phase 2
Municipal PO credit flow	Manual PO upload + admin approval vs PFMS API integration	Manual PO upload for Phase 1; PFMS in Phase 3	Phase 1
Price recommendation AI	Rule-based median vs regression vs GPT-assisted pricing	Rule-based median first; switch to regression at 1,000+ orders	Phase 3